OAK INTERNATIONAL SCHOOL

MYP 5 MATHEMATICS — MID-TERM TEST

Bilingual Excellence in Education & Leadership

DURATION	30 minutes
TOTAL MARKS	20
CALCULATOR	Not permitted

Name: _____ **Class:** _____ **Date:** _____

Answer ALL questions on this paper. Show all working clearly — marks are awarded for method, justification and clear communication. Write final answers on the dotted lines. Give non-exact decimal answers correct to 2 decimal places. Aim for about 90 seconds per mark.

TASK 1 — RECOGNISING AND UNDERSTANDING SEQUENCES — Criterion A • 4 MARKS

Two sequences are shown below.

P: 5, 11, 17, 23, … **Q:** 2, 6, 18, 54, …

(a) For sequence P, write down the common difference.

.....

[1 mark]

(b) Determine whether sequence Q is **linear**. Justify your answer by considering the differences between consecutive terms.

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.....

[2 marks]

(c) Write down the next term of Q.

.....

[1 mark]

TASK 2 — ARITHMETIC SEQUENCES AND GENERAL TERMS — Criterion A • 6 MARKS

The first four terms of an arithmetic sequence **U** are:

U: 23, 19, 15, 11, …

(a) Find u_{25}, the 25th term of U.

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[2 marks]

(b) A different arithmetic sequence **V** has fourth term 19 and common difference –3. Find the first term of V, and write down the first four terms of V.

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[2 marks]

(c) Explain, with clear working, why –60 cannot be a term of U.

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[2 marks]

TASK 3 — APPLYING SEQUENCES: THEATRE SEATING — Criterion D • 6 MARKS

A theatre in Cotonou is hosting the school arts showcase. Row 1 (the front row) contains 12 seats, and each row behind it contains exactly 3 more seats than the row in front of it. There are 20 rows in total.

(a) Show that the number of seats in row n is $u_n = 3n + 9$.

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[2 marks]

(b) Find the number of seats in row 15.

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[1 mark]

(c) Calculate the total number of seats in the theatre.

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[2 marks]

(d) A full-school performance requires 795 seats. Determine whether **one performance** in this theatre is enough for the whole school. Justify your answer.

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[1 mark]

TASK 4 — REASONING CHALLENGE: TWO PATTERNS — Criterion A • 4 MARKS

Pattern X begins at 100 and decreases by 8 each time. Pattern Y begins at 10 and increases by 12 each time. The patterns are compared term by term (term 1 with term 1, term 2 with term 2, and so on). A student claims: *"From the 5th term onwards, Y is always greater than X."*

(a) Write down a formula for the nth term of each pattern.

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[1 mark]

(b) Investigate the student's claim. Find, with full justification, the first paired term for which Y is greater than X.

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[3 marks]